

POWER BI PROJECT

INTRODUCTION:

Power BI is a powerful business intelligence tool developed by Microsoft that allows users to visualize, analyze, and share insights from data. In today's data-driven world, organizations rely on clear and interactive reports to make better decisions. Power BI makes it possible to transform raw data into visually appealing dashboards that provide actionable insights.

WHAT IS POWER BI?

Power BI is a data visualization and business analytics tool that connects to various data sources and helps users create interactive reports and dashboards. It combines data preparation, visualization, and sharing in one platform.

POWER BI PROCESS:

- **Data Connectivity** – Importing or connecting live data from sources like Excel, SQL, or cloud platforms.
- **Data Transformation** – Cleaning, shaping, and modeling data using Power Query and DAX.
- **Data Visualization** – Creating interactive charts, graphs, maps, and KPI cards.
- **Dashboard Creation** – Building user-friendly dashboards that highlight key business metrics.
- **Data Sharing** – Publishing reports to the Power BI Service for collaboration and decision-making.

WHAT IS POWER QUERY EDITOR?

Power Query Editor is a data transformation tool in Power BI (also available in Excel). It allows users to connect, clean, and shape data before loading it into Power BI for analysis and visualization. It provides an easy-to-use interface with point-and-click options as well as an underlying M language (Power Query Formula Language) for advanced transformations.

Key Features of Power Query Editor:

- **Data Import** – Connect to multiple sources like Excel, SQL, Web APIs, CSV, etc.
- **Data Cleaning** – Remove duplicates, handle missing values, filter out irrelevant rows, and correct errors.

- **Data Transformation** – Split columns, merge tables, pivot/unpivot, change data types, and create custom columns.
- **Data Shaping** – Reorganize data structures to make them analysis-ready. Steps – Every transformation is recorded as a step, making the process transparent and reversible.
- **Automation** – Once defined, the same query can be refreshed automatically with updated data.

“BANK TRANSACTION DASHBOARD”

INTRODUCTION:

This dataset contains over 5,000 banking transactions collected for status analysis and location-based insights. Each row represents a unique transaction with details like amount, date, location, and account type. The data is used to detect the transaction status, visualize city-wise activity, and support decision-making for banking operations.

COLUMNS DETAILS – BANK TRANSACTION

- **Transaction ID** - Unique ID for each transaction.
- **Customer ID** - Identifier for the customer.
- **Account Type** - Type of account (e.g., Credit, Savings).
- **Transaction Type** - Nature of transaction (Deposit, Transfer, etc.).
- **Amount** - Transaction amount in ₹.
- **Date** - Date of transaction.
- **Time** - Time of transaction.
- **Location** - City or branch where transaction occurred.
- **Channel** - Mode of transaction (Branch, ATM, Online).
- **Status** - Current status (Pending, Completed, Failed).

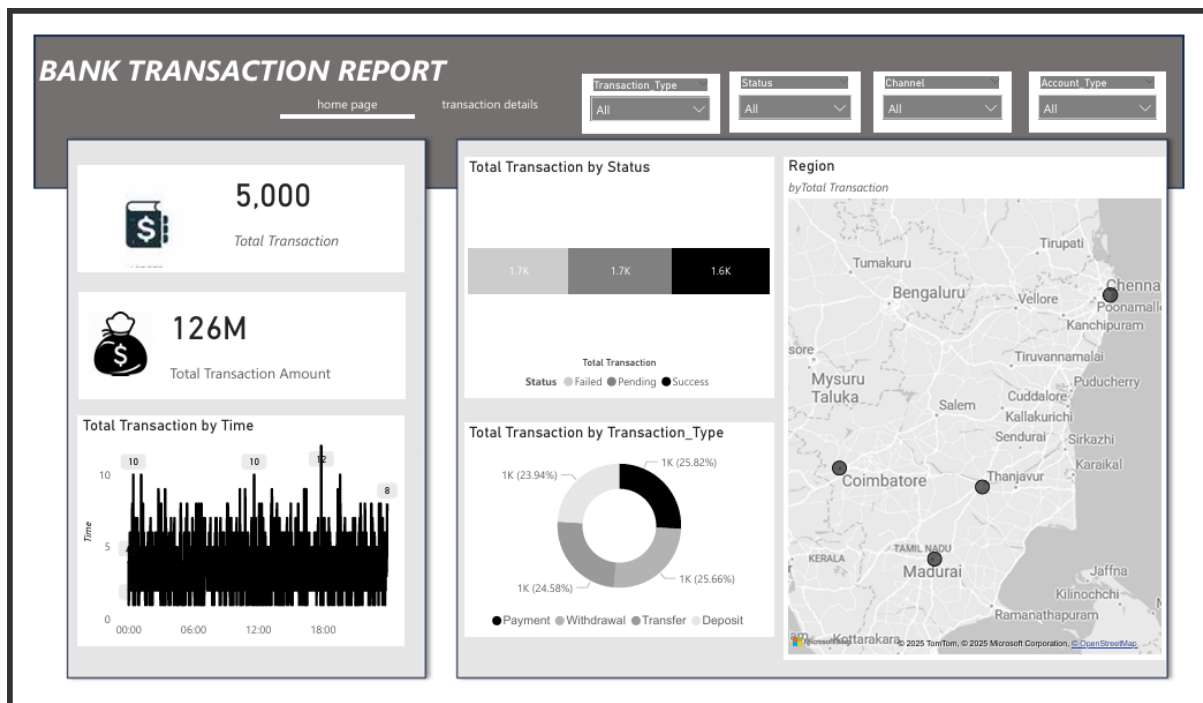
DATA CLEANING & PREPARATION NOTE:

The dataset was directly imported into Power BI from Kaggle. Normally, in supply chain analytics, the Power Query Editor is used to clean and preprocess the dataset (e.g., handling missing values, removing duplicates, renaming columns, fixing data types).

However, in this case:

- The dataset contained no missing values.
- There were no duplicates.
- Column data types were correctly assigned (numeric, categorical, etc.).
- Values were already in a clean and consistent format.

DASHBOARD:



STATUS DETECTING AND DETAILS: